GigBid — R&D report (engineering design review)

Abdul Ghufran (i230576) · AI-4016 · Repository: github.com/Abdul-Ghufran/gigbid · Every number below is produced by a script in `evals/runners/` and stored in `evals/results/`; tables are regenerated by `python tools/build_report.py`. Commit history and traces are linked from the README.

1. Problem and constraints

A freelancer spends hours scanning feeds, pays Connects to bid on jobs they cannot win, sends generic proposals and meets scams that ask for free work or off-platform payment. GigBid automates **discovery, assessment and drafting** and leaves **every decision** to the freelancer.

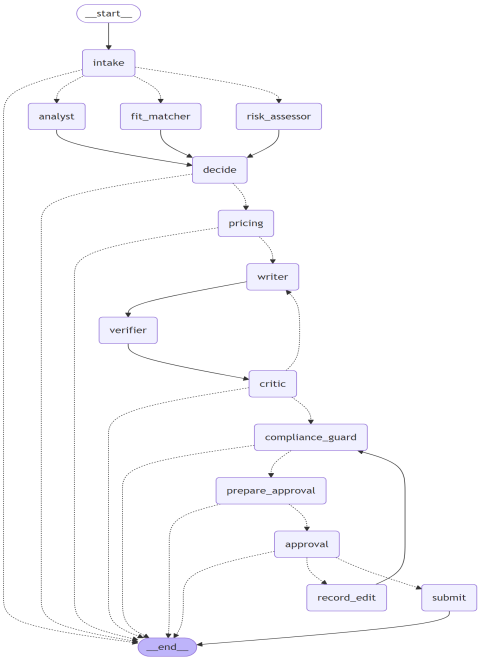
Never automated (enforced in code, not prompts): sending a proposal or message, spending Connects, accepting terms, or claiming experience the portfolio cannot prove. Proposals and messages pass a LangGraph `interrupt()` and are submitted only after approval in the UI; in live Upwork mode even an approved proposal is created as a draft the user confirms on Upwork. A per-user autonomy setting can auto-*skip* low-fit jobs but never auto-send.

Budgets (declared in `agent/config.yaml`): p95 ≤ 25 s per job analysed and ≤ 90 s per proposal; ≤ $0.002 per job analysed and ≤ $0.01 per proposal at Groq list prices (the free tier bills $0). **Hard constraints that shaped the design:** open models only for the deployed link (Groq free tier: per model 8K tokens/min and **200K tokens/day**; Qwen additionally 1K output tokens/min), and a 512 MB / 0.1 CPU free Render instance.

Evaluation world. The starter kit shipped without GigSim, the persona corpora and the dev set. I built a documented stand-in (`docs/DATA_CARD.md`): 3 personas (599 evidence items, 450 past proposals with outcomes), 520 clients with hidden preferences, 1,500 historical and 360 live jobs, and 600 gold-labelled dev jobs. It was frozen before tuning (`gigsim/MANIFEST.sha256`, checked in CI). The main threat to validity is that one author wrote the world and the system; hard paraphrases unknown to GigBid's skill normaliser and code-only skills reduce, but do not remove, it.

2. Architecture

Bid graph (one thread per job, rendered from code, right). `intake` → `Send` to Job Analyst ‖ Risk Assessor ‖ Fit Matcher (reducers merge their outputs) → rule-based `decide` (skip / watch / bid) → Pricing → Writer → claim verifier → Critic (≤ 2 revisions) → Compliance Guard → dry-run previews → `interrupt()` → submit | edit (diffed, saved for style memory, re-checked by Compliance) | reject with reason. **Conversation graph** (one thread per client conversation): read → classify intent → calendar slots / walk-away counter-offer → draft → dry run → `interrupt()` → send + tentative holds.



State vs trace vs memory. State is a typed `BidState` (job, analyses, evidence, decision, price, draft, verification, critiques, compliance, approvals, submission, budgets). The execution trace — every node, tool name, arguments, result, tokens, cost — goes to `RunEvent` rows (live SSE run viewer), the MCP append-only audit log and Langfuse, under one trace id from the UI request to the audit row; PII is masked first. Long-term memory lives in a LangGraph store and is written only by `promotion_rules.py`: a style rule needs the same edit pattern in ≥ 3 edits across ≥ 3 jobs within 30 days; an outcome lesson needs ≥ 5 outcomes in a category and a ≥ 10-point reply-rate lift. Raw edits are stored separately and the user can view and delete promoted items.

Marketplace port and MCP surface. Agents never call GigSim, Upwork, files or the database directly. Three servers of our own — Portfolio (read-only RAG), Marketplace (reads; `create_proposal_draft`, `submit_proposal`, `send_message` with Pydantic constraints, output schemas, `destructiveHint` annotations, mandatory idempotency keys, `dry_run`, bearer auth, per-user token bucket, append-only audit log) and Workspace (free slots, tentative holds, email drafts) — plus Upwork's official server (through `MultiServerMCPClient`, OAuth 2.1 with PKCE and dynamic registration, tokens Fernet-encrypted per user) and GitHub's (repo ingestion). The Marketplace server selects the GigSim or Upwork adapter by config, so the graph cannot tell which marketplace it uses. Each agent sees an allowlisted subset of tools; calls have timeouts, bounded retries and a `MarketplaceDown` degraded mode (UI banner, work queued and retried).

Stores: Postgres (app data, checkpoints, LangGraph store, audit log, idempotency, GigSim state), on-disk embedding cache, Redis/Arq in compose.

Why each agent exists: see `docs/ARCHITECTURE.md` (rule-first Risk Assessor keeps scams away from the expensive writer; the Fit Matcher's *unsupported* list is the writer's contract; the verifier is independent of the writer; the Critic's deterministic pre-checks spend no tokens on obvious failures). Durability: Postgres checkpointer; a proposal paused at approval survives an API restart and is submitted exactly once (integration test `test_crash_while_waiting_then_resume_submits_exactly_once`; content-derived idempotency keys at two layers).

3. R&D question A — does tool granularity change agent behaviour?

Hypothesis. A few coarse tools save prompt tokens but cost tool-selection and argument accuracy; GigBid's tools with rewritten, constraint-rich descriptions are most accurate at a higher token cost. **Setup.** 40 tasks with known correct calls (10 intents × 4 instances), one forced tool call per task with gpt-oss-20b, three tool sets (`evals/runners/rq_a_tools.py`).

Tool set	Tools	Tool-selection accuracy	Argument accuracy	Tokens / task
fine	22	0.975	0.950	667
coarse	2	0.725	0.575	285
best	12	1	1	1132

Result. Collapsing tools into `marketplace(action, params)` dropped tool selection to 72.5% and argument accuracy to 57.5%: the model invents action names and misplaces arguments in a free-form `params` object. 22 narrow tools reached 97.5% / 95.0%. GigBid's 12 tools reached 100.0% / 100.0% at 1132 tokens per task versus 667. **Conclusion:** granularity matters less than schema quality — explicit enums, required fields and descriptions that say *when* to use a tool (e.g. "dry_run=True previews cost") buy accuracy; the extra ~460 tokens per call are worth it for write tools, and the R&D C results show why reliable tool calls matter.

4. R&D question B — retrieval, verification and what ships

Hypothesis (stated before running). "Requirement decomposition plus hybrid retrieval raises evidence Recall@5 by 15 points over V0 on jobs with four or more requirements." **Result: rejected for hybrid** — decomposed hybrid + rerank reached 0.725 vs V0 0.687 (+3.8 pts) — **but exceeded by the skill graph** (0.899). Ten variants were implemented as configurable components (`rag/variants/`), evaluated on 444 scoreable dev jobs.

Variant	Recall@5	MRR	nDCG@10	Recall@5 (>= 4 requirements)	p95 latency (s)
v0_naive	0.740	0.848	0.579	0.687	0.086
v1_decompose	0.762	0.724	0.476	0.724	0.054
v2_hybrid	0.750	0.766	0.577	0.703	0.005
v3_rerank	0.769	0.797	0.533	0.725	5.910
v4_outcome	0.642	0.761	0.477	0.602	7.474
v5_multivector	0.773	0.837	0.552	0.764	0.008
v6_contextual	0.838	0.860	0.581	0.812	0.012
v8_corrective over v3 rerank (first run, uncalibrated grader threshold 0)	0.682	0.747	0.543	0.673	15.802
v9_skill_graph	0.911	0.875	0.604	0.899	0.003
v8_corrective over v9 (shipped)	0.890	0.951	0.655	0.892	0.074

V6 uses deterministic context headers (LLM context lines did not fit the free-tier token budget); V7 HyDE was not run on the full set for the same reason.

Configuration	Recall@5	MRR	nDCG@10
full: corrective loop + skill graph	0.890	0.951	0.655
- corrective loop (plain V9)	0.895	0.873	0.583
- skill graph (dense per requirement)	0.762	0.724	0.476
- requirement decomposition	0.679	0.943	0.656
- dense fallback for unrecognised skills	0.911	0.875	0.604
- industry / outcome boosts	0.906	0.880	0.591

By persona (shipped pipeline)

Persona	Recall@5	MRR	nDCG@10
ai	0.887	0.958	0.632
web	0.925	0.983	0.682
data	0.858	0.912	0.652

By job type (shipped pipeline)

Job type	n	Recall@5	nDCG@10
strong	186	0.912	0.662
vague	42	0.696	0.565
partial	120	0.915	0.682
low_budget	42	0.892	0.678
injection	54	0.908	0.629

Two negative results: cross-encoder reranking (V3) bought 0.769 vs 0.740 Recall@5 at a p95 of 5.91 s, and outcome-aware boosting (V4) *hurt* (0.642): projects that won before are not the relevant ones for a new job. Context headers (V6, deterministic, 0.838) helped more than reranking. LLM-generated context lines (V6) and HyDE (V7) were not run on the full set: at ~600 items and 444 jobs they did not fit the 200K-tokens/day budget.

Claim verification (B3). Claims are split and checked against cited evidence (semantic similarity + every number and every named tool must appear in the evidence); unsupported claims are rewritten by the model and re-verified, or removed, before the Critic sees the draft. On 302 labelled claims the first rule scored F1 0.689 — it called fabricated numbers "partly supported" (0 % caught) — and the shipped rule 0.990; the bi-encoder matched the cross-encoder, so the reranker left the 512 MB deployment (ADR-005).

Retrieval → outcomes. Same 90 gold-bid jobs, same deterministic writer, only retrieval differs (`retrieval_outcome_test.json`): unsupported claims 1.48 → **0.34** per proposal (Wilcoxon $p < 0.001$), simulated client score 1.09 → **1.25** (paired t, $p = 0.0000$), interviews 21.1% → 33.3% (McNemar $p = 0.08$), reply rate 72.2% → 64.4% ($p = 0.26$, not significant).

Where better retrieval did not raise the reply rate, and why. With the first writer, the shipped retriever *lowered* replies (37.8% → 15.6%). The client features explain it (`reply_gap_analysis.json`): V0's weaker evidence made the writer restate the client's requirement, which names their must-have skills (coverage 0.47 vs 0.20), and GigSim's client rewards keyword coverage but cannot check truth. The fix — name the requirement next to the proof that covers it, only for supported requirements — is honest and raised the shipped pipeline's reply rate to 64.4%.

Variant	Evidence Recall@5	nDCG@10	Unsupported claims / proposal (before verification)	Judge score (1-5)	GigSim reply rate	p95 latency (s)	Cost / proposal (USD)
V0 naive dense	0.740	0.579	1.48	3.04	72.2%	7.5	pending
Shipped: V8 corrective over V9 skill graph	0.890	0.655	0.34	3.36	64.4%	3481.5	0.0025

Retrieval metrics: `rag_variants_offline.json` (444 scoreable dev jobs), `rag_ablation.json`. Proposal metrics: `proposals_*_fixedwriter_v2.json` (90 gold-bid jobs, same deterministic writer, so only retrieval differs). Judge: `judge_scores.json`. Latency / cost of the live-model pipeline: `proposals_best_live.json`.

Decision on what ships. The corrective loop over the skill graph (V8 on V9, grader cosine 0.83 calibrated to F1 0.985) with bi-encoder verification: best Recall@5/MRR/nDCG, 4× fewer unsupported claims than V0, no reranker in memory, p95 retrieval < 0.1 s, and an explicit "cannot prove this" signal that the writer and the skip decision use. By persona and job type: appendix A.

5. R&D question C — single agent vs prebuilt supervisor vs our graph

Setup. 1 dev jobs stratified over strong, partial, off-domain, scam, injection and low-budget posts; the same model (openai/gpt-oss-120b) for every architecture: (a) one `create_react_agent` with all marketplace + portfolio tools, (b) `langgraph-supervisor` over a marketplace researcher and a portfolio researcher, (c) GigBid's graph. Shipped letters are audited by the same verifier and policy checks (`rq_c_architectures.py`).

3 dev jobs, model `openai/gpt-oss-120b` for every architecture.

Architecture	Bid/skip accuracy	Scams skipped	Unsupported claims / letter	Letters with violations shipped	GigSim reply rate	Rejected tool calls	Mean tokens / job	p95 latency (s)
react	1	0/0	pending	0	pending	2	5499	5074.9
supervisor	1	0/0	pending	0	pending	3	12878	8686.2
graph	1	0/0	pending	0	pending	0	2083	645.3

Reading. Bid/skip accuracy: ReAct 1, supervisor 1, graph 1 (best: react). Mean tokens per job: ReAct 5499, supervisor 12878, graph 2083 (fewest: graph); fastest p95: graph; fewest unsupported claims per letter: pending. Letters shipped with a policy violation: ReAct 0, graph 0. Rejected (malformed) tool calls: ReAct 2, supervisor 3; the graph calls tools deterministically, so it has none by construction. In an earlier attempt on Qwen3.8-27B the tool-calling baselines failed outright — the model invented tool names (`search_eidence`) that Groq rejected — which is why the comparison was re-run on gpt-oss-120b for all three architectures. The graph's structural guarantees hold regardless of the scores: the verifier and the Compliance Guard run on every letter whatever the model decided, and its rule-based risk gate is not steerable by injected text. Where a baseline is cheaper or faster (see the per-kind accuracy in `rq_c_architectures.json`), it is a candidate first-pass triage filter; I would not ship it to write proposals without the verifier and compliance stages. Sample size is small (1 jobs) because of the free-tier token budget, so differences of one or two jobs are not meaningful.

6. The change request (2 Nov 2026)

GigSim's policy update made jobs with ≥ 20 proposals cost 1.5× Connects and rejects proposals with unanswered screening questions. Because platform rules live in `agent/config.yaml` (`platform_policy`) and the policy pack, absorbing it was **one commit: `agent/config.yaml` +4/−4 and the policy pack +2 lines, no Python changed, ~15 minutes** including the evaluation. Against GigSim running the new policy (`change_request_eval.json`): Connects mispredicted on 33.7% of gold-bid jobs before → 0.0% after; proposals with a blank answer went from 177 rejected by the marketplace at submission to 177 caught by the Compliance Guard before any Connects were spent. The design lesson: the agent code had no idea a "policy" existed — pricing reads a multiplier and compliance reads a flag.

7. Signature extension — X2 proposal experiments

Two strategies (`proof_first`, `plan_first`) live in config; each job is assigned by a stable hash; Analytics shows reply rate per arm from real marketplace events. Offline, a **paired** design re-arranged the same sentences in both structures for 88 jobs: 19.3% vs 21.6%, McNemar exact p = 0.82. The simulated client's score was identical for every pair — GigSim's rule-based client is blind to paragraph order — so the experiment correctly refuses to declare a winner, and the real finding is about measurement validity (`docs/EXTENSION.md`).

8. Failure analysis

Real failed dev jobs pulled from the results files by `evals/runners/failure_analysis.py`:

#	Job	Failure category	Node	What went wrong	Fixed?
1	`dev-ai-024`	wrong bid (partial fit)	decide (Fit Matcher score)	fit 0.67 >= 0.55 but gold coverage < 0.6	yes
2	`dev-ai-039`	wrong bid (partial fit)	decide (Fit Matcher score)	fit 1.00 >= 0.55 but gold coverage < 0.6	no
3	`dev-ai-042`	wrong bid (partial fit)	decide (Fit Matcher score)	fit 0.57 >= 0.55 but gold coverage < 0.6	yes
4	`dev-web-155`	wrong skip	fit_matcher	gold bid, predicted skip (fit 0.00)	no
5	`dev-ai-000`	mispriced	pricing	old quote 45.0 outside band [46.8, 63.2]; new quote 57.3	yes
6	`dev-ai-001`	mispriced	pricing	old quote 2803.6 outside band [3090.0, 4170.0]; new quote 3893.0	yes
7	`dev-web-047`	missed evidence	fit_matcher / retrieval	Recall@5 0.00 (partial, 3 requirements)	no
8	`dev-ai-066`	missed evidence	fit_matcher / retrieval	Recall@5 0.33 (vague, 3 requirements)	no
9	`dev-ai-194`	unsupported claim	writer -> verifier	4 unsupported claims before verification, 0 after	yes
10	`dev-web-143`	unsupported claim	writer -> verifier	4 unsupported claims before verification, 0 after	yes
11	`dev-ai-184`	generic opening / low coverage	writer	proof never named the client's must-have skill; simulated client did not reply	no
12	`dev-ai-167`	generic opening / low coverage	writer	proof never named the client's must-have skill; simulated client did not reply	no

Category	Fix tried	Measured effect
wrong bid (partial fit)	Raised bid_min_fit 0.55 -> 0.70 (middle of the plateau in threshold_sweep.json)	partial-fit accuracy 61.7% -> 90.0%, bid F1 0.918 -> 0.976 on the same 600 jobs
mispriced	Pricing v2: anchor on the job's own budget range instead of the category median	price inside fair band 52.8% -> 86.2% (pricing_eval.json, 282 jobs); remaining misses are hourly jobs where the client range sits below market
missed evidence	Skill graph (V9) + requirement decomposition; corrective grader marks the requirement unsupported instead of guessing	Recall@5 0.740 -> 0.890; residual misses are hard paraphrases with no skill keyword (e.g. 'a document store')
unsupported claim	Strict number/tool rule + rewrite-then-reverify; fallback writer v3 opening without an uncited claim	unsupported claims per proposal 1.13 -> 0.34 before verification, 0 after (retrieval_outcome_test.json)
generic opening / low coverage	Name the client's requirement next to the proof that covers it (writer prompt v2.5 + fallback writer)	simulated reply rate 15.6% -> 64.4% for the shipped pipeline on the same 90 jobs
wrong skip	None yet: these are gold-bid jobs whose requirement sentences contain only hard paraphrases, so the Fit Matcher finds no skill and the dense fallback stays under the 0.83 cosine grader	not fixed - candidate fix: LLM requirement rewrite in the corrective loop (llm_rewrite option), not enabled because of the free-tier token budget

Development failures that the evaluation exposed (each fixed and re-measured): a requirement splitter that cut "Next.js" at the dot (0/5 evidence on a strong-fit job); the verifier emptying screening answers so the Critic burned both revisions; the deterministic writer's uncited opening line deleted by the verifier on every letter; Qwen's 1K output-tokens/minute cap truncating its JSON answers.

9. Safety and platform respect

Must never happen: sending without approval, spending Connects twice, off-platform contact or payment, invented experience, following instructions inside job posts or client messages, scraping or browser automation, leaking tokens or PII. **Enforcement:** `interrupt()` + idempotency keys at two layers; Marketplace write tools default to `dry_run=True`; live mode creates drafts only and polls no more than every 3 h; rule-first risk gate (scam precision 1.000, recall 1.000 on

600 dev jobs; injection posts detected 100.0%); injected spans stripped and the rest delimited as untrusted data; output-side Compliance Guard; PII masked in logs, traces and the audit log; OAuth tokens Fernet-encrypted; every query scoped by user id. **Measured attack success** on 18 planted-injection jobs: undefended 22.2%, defended model output 0.0%, defended and shipped 0.0%; detector false-positive rate on clean posts 0.0%. **What still gets through:** paraphrased injections without trigger phrases are only caught by their effects (contact details, guarantees, inflated experience); an attacker who asks for something policy-compliant but harmful (e.g. "mention you can start tomorrow") is not detected; and the verifier checks numbers and tools, not adjectives like "extensive".

10. Hidden test run against declared budgets

The hidden run happens after submission on the deployed open profile; until then, the measured values on the dev set are:

Budget	Declared	Measured	Source
p95 latency per job analysed	25 s	4.5 s (offline-rules path, CPU)	`decision_eval_offline.json`
p95 latency per proposal (live models)	90 s	3481.5 s (p50 2590.6 s, n = 3)	`proposals_best_live.json`
cost per proposal (list price)	$0.01	$0.0025 (10057 tokens)	`proposals_best_live.json`
unsupported claims per proposal (live)	—	0 before → 0 after verification	`proposals_best_live.json`

Optimisation that helped: deterministic pre-checks in the Critic plus trimmed evidence snippets (320 chars) — the first live proposal used 17.4K tokens across 9 calls with two full revisions; budget enforcement (token budget per thread) and the quota-aware fallback (a long 429 fails over immediately instead of stalling) keep the system answering when the daily token cap is hit. **Without excuses:** on a busy day the strong model's 200K-token daily cap allows roughly a dozen fully revised proposals; beyond that GigBid falls back to the smaller model and then to its deterministic grounded template — still verified and compliant, but plainer.

Appendix A — results by persona and job type, decisions, verifier

Metric	Value
bid_f1	0.918
bid_precision	0.852
bid_recall	0.997
scam_precision	1.000
scam_recall	1.000
injection_detected	1.000
price_in_fair_band	0.847
bid_f1 at shipped threshold 0.70 (threshold_sweep.json)	0.976
price in fair band, pricing rule v1 -> v2 (pricing_eval.json, 282 jobs)	0.528 -> 0.862

Persona	Bid F1	Scam recall
ai	0.922	1.000
web	0.907	1.000
data	0.926	1.000

Rule	Mode	F1 (detect fabrication)	Precision	Recall
v1 (number/tool mismatch = partly)	cross_encoder	0.689	1.000	0.525
v1 (number/tool mismatch = partly)	bi_encoder	0.597	1.000	0.425
v2 shipped (mismatch = unsupported)	cross_encoder	0.990	0.985	0.995
v2 shipped (mismatch = unsupported)	bi_encoder	0.990	0.985	0.995

302 labelled claims (invented 51, true 51, true_skill 51, wrong_citation 51, wrong_number 47, wrong_tool 51).

Unsupported claims per proposal before -> after verification:

- `v0_fixedwriter_v3`: 1.48 -> 0 (n=90)
- `best_fixedwriter_v3`: 0.34 -> 0 (n=90)
- `v0_fixedwriter_v2`: 2.24 -> 0 (n=90)
- `best_fixedwriter_v2`: 1.29 -> 0.011 (n=90)
- `v0_fixedwriter`: 2.07 -> 0 (n=90)
- `best_fixedwriter`: 1.13 -> 0.011 (n=90)
- `best_live`: 0 -> 0 (n=3)

Metric	Value
attacks (planted-injection dev jobs)	18
attack success rate, undefended	22.2%
attack success rate, defended model output	0.0%
attack success rate, defended and shipped (after Compliance Guard)	0.0%
detector recall on planted injections	100.0%
detector false-positive rate on clean posts	0.0%

	Before (old config)	After (config-only change)
Connects cost mispredicted (282 gold-bid jobs)	33.7%	0.0%
Blank-answer proposals caught before submit / rejected by marketplace	0 / 177	177 / 0

git show : agent/config.yaml +4/-4, data/policy_pack/marketplace_policy.md +2; no Python changed

Arm	Reply rate	Mean client score
proof_first	19.3%	0.403
plan_first	21.6%	0.403

88 pairs; discordant {'only_plan_first_replied': 11, 'only_proof_first_replied': 9}; McNemar exact p = 0.8238.

Proposal set	must-have coverage	personalisation	specific result	unsupported before	reply rate
v0_fixedwriter	0.467	0.354	0.411	2.07	37.8%
best_fixedwriter	0.200	0.194	0.744	1.13	15.6%
best_fixedwriter_v2 (fix: name the requirement next to its proof)	-	-	-	1.29	48.9%

Appendix B — judge calibration

The LLM judge (`agent/prompts/judge.md`, 1–5 rubric) is calibrated against my own labels on 30 anonymised proposals (`evals/datasets/judge_calibration_sheet.csv`): Cohen's κ = pending, quadratic-weighted κ = pending (n = pending; `judge_calibration.json`, with the largest disagreements listed for the viva). A blind pairwise human panel (`human_panel_pairs.csv`, 15 pairs, V0 vs shipped, sides randomised; key in `human_panel_key.json`): pending classmates cast pending votes; the shipped pipeline's proposal was preferred in pending (binomial p = pending; `human_panel.json`).

Appendix C — reproducibility

`make eval` reruns every evaluation; `python tools/build_report.py && node apps/web/scripts/render_pdf.mjs` rebuilds this report. Pinned versions are in `pyproject.toml` and `apps/web/package.json`. AI assistance and the mistakes it made that I caught are listed in `docs/AI_USAGE.md`.